Statistical methods for computer science

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Basic Concepts

Course outline

- Basic concepts
- Descriptive statistics
- Introduction to probability
- Introduction to random variables
- Discrete probability distributions
- Continuous probability distributions
- Sampling distributions
- Statistical inference: estimation
- Statistical inference: testing
- Linear model: estimation, inference, diagnostics, aplications

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Exam
Written test with excercises

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Statistical science: what for?

We developed a brand new video game, called "Star Game", about a new universe and a group of explorers seeking a new Earth. Before commercializing it, we perform a market research by sending a trial version of the game and a questionnaire to 1000 people

Results:

- 82% of the respondents liked the video game
- The game scored an average rating of 3.5 stars out of 5

Can we say that Star Game will be successful?

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Results:

- 82% of the respondents liked the video game
- The game scored an average rating of 3.5 stars out of 5

Can we say that Star Game will be successful? No/Not yet

Statistical science: what for?

Science of developing and applying methods for collecting, analyzing and interpreting data.

- **Design:** planning how to gather relevant data

  e.g., how to select the people to interview, how to construct the questionnaire
- **Description:** summarizing the data (descriptive statistics)

  e.g., average score obtained by the game in the *sample* of 1000 respondents
- **Inference:** estimations and predictions based on the data (statistical inferences)

  e.g., average score and related uncertainty in the *population* if we imagine to administer the questionnaire to all Italian young adults

Population vs sample

The entities under study are called **subjects** (e.g., people, families, counties etc.)

The population is the total set of subjects of interest, a sample is the set of subjects from the population for which data are available. A parameter is a numerical summary of a population, a sample statistic describes the given sample

Population vs sample

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- Italian young adults (population) vs the interviewed 1000 young adults (sample)
- The average score of 3.5 is a sample statistic, whereas the average score that the game would obtain in the whole population of Italian young adults is a parameter

Population vs sample

Researchers' primary interest is the values of parameters rather than sample statistics

- We know so far that among the 1000 sampled young adults, 82% of them like the new video game, but we are more interested in determining the percentage of young adults that will like Star Game in the population of all Italian young adults

The process of drawing conclusions about a population based on data collected from a sample is called **inference**

The first step for a reliable inference is the choice of the sampling method

Simple random sampling

Probabilistic sampling technique in which every member of the population has the same probability of being selected

- *The game developers randomly select 1,000 people from all the 100,000 students attending the 5-th year of Italian high school. Each of them has a probability of* $1,000/100,000 = 1\%$ *of being sampled*

Pros

Very simple, fair, tends to yield a sample that resembles the population reducing the chance of inaccurate inferences

Cons

Expensive, there is a possibility of under-representing or over-representing certain subgroups within the population

Simple random sampling

Exercise

We want to sample 3 subjects from a population of size 12 using simple random sampling: 1) what is the probability P of each subject to be selected? 2) how many possible samples there exist?

1. $P = 3/12 = 0.25$
2. We can select 3 people out of 12 in $\binom{12}{3} = \frac{12!}{3!9!} = C_{12,3} = 220$ possible ways (i.e., combinations without repetitions)

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Given a population of size N, the number of possible samples of size $n \leq N$ is

$$N_s = \binom{N}{n} = \frac{N!}{n!(N-n)!}$$

Types of Variables

Design phase:

- ✓ Definition of target population
- ✓ Choice of the sampling scheme

Now we need to determine what kind of information we need to answer our research questions and how to measure them.

Example

Our goal is understanding if our target population (Italian young adults) will like Star Game or not. How can we measure "liking"?

- Ask a direct question ("Did you like the trial version?") and collect yes-no answers
- Ask the subjects to quantify how much they liked the game on a scale from 1 to 5
- Measure the actual time that each subject spent on the game and/or the number of games they played

Types of Variables

In other words, what variables do we want to collect?

> *A variable is a characteristic that can vary among subjects in a sample or population*

Variables can be classified according to the type of values that they can take as:

- Categorical (or qualitative)
    - Nominal
    - Ordinal
- Numerical (or quantitative)
    - Discrete
    - Continuous

Categorical variables

The measurement scale is a set of categories. They can be further classified in:

Nominal
Unordered categories. Variables with only two categories are called *binary*

Examples: marital status (*married*, *single*, *divorced*, *widowed*), whether employed (*yes*, *no*), employment status (*employed*, *unemployed*, *self-employed*, *retired*)

Ordinal
Ordered categories (natural ordering of values)

Examples: level of agreement with a statement (*strongly agree*, *agree*, *neither agree nor disagree disagree*, *strongly disagree*), perceived happiness (*unhappy*, *happy*, *very happy*)

Numerical variables

According to the number of values in the measurement scale, numerical variables can be further classified in:

Discrete

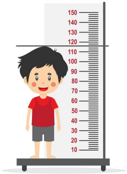
It takes a set of distinct, separate values

Examples: number of students in a class, number of cars in a parking lot

Continuous

It takes an infinite continuum of real number values

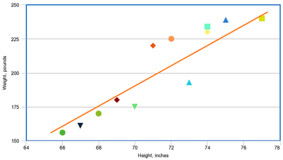
Examples: age, weight, height, temperature, distance



Response vs explanatory variables

An association between two variables describes how the values or measurements of one variable are related to the values of another variable.

- Positive
- Negative
- No association



Based on their association, we can distinguish between:

Response variable
Also called *outcome* or *dependent* variable

Explanatory variable(s)
Also called *independent* variable(s)

Exercise

What is the type of the following variables measuring "liking"?

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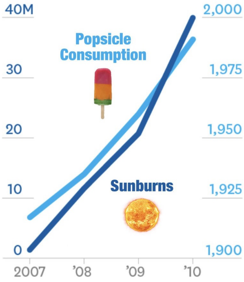
- Measure the actual time that each subject spent on the game and/or the number of games they played: numerical continuous

We decide to measure "liking" with the ordinal variable and to measure the willingness to purchase asking weather they would buy the game or not. We also collect further information from survey participants: age, sex, educational attainment, employment status, annual income, city of residence. Which ones of these are the (in)dependent variables?

Association vs causation



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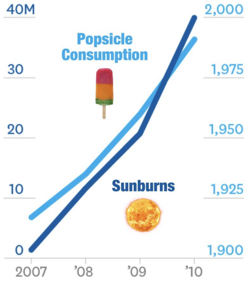

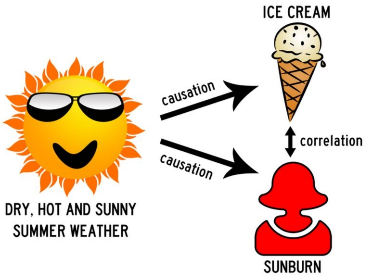

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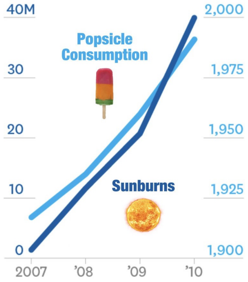

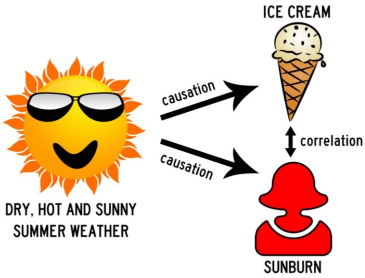

Association does not imply causation!

How to establish cause and effect?

Suppose that based on the results of the survey (82% of the respondents liked the video game) we released "Star Game" but the sales are lower than expected.

We are planning to give some incentive to boost sales (e.g., 20% discount on the next purchase) but before doing that, we want to evaluate whether this incentive will be effective or not. In other words: *assigning this incentive will yield an increase in sales?*

This is a causal question

How to establish cause and effect?

Experimental study

We randomly sample some people visualizing the description of the game on the online shop (simple or stratified sampling) and assign them to two groups. One group will be given the incentive (treated group) whereas the other group will be given no incentive (control group)

Easier to estimate the causal effect of the incentive (randomization balances the two groups, any difference can be attributed to the incentive)

Observational study

We recover daily sales data of all our video games in the past 2 years. In the database there is a binary variable taking value 1 when a game in a certain day was associated to the 20% incentive and 0 otherwise. Stratifying based on this variable we still obtain a treated and a control group

More challenging (we have no knowledge of the assignment mechanism, the groups may be unbalanced) but **possible**

Where are we?

Statistical science is the science of developing and applying methods for collecting, analyzing and interpreting data.

Design the experiment:

- ✓ Definition of target population
- ✓ Choice of the sampling scheme
- ✓ Decision on the information to collect and how to measure them

Description of results: next lecture!